St. Philomena's College (Autonomous), Mysore			
PG Department of Studies and Research in Chemistry			
Question Bank			
First Year – First Semester			
Course Title (Paper Title): Inorganic Chemistry-I **Q.P Code: 85122**			
Unit/ Module	**SI. No**	**Questions**	**Marks**
1	1.	Most common acids behave as bases in liquid H_2SO_4. Suggest an explanation for	02
1	2.	What are super acids? How hydrocarbons behaves in super acidic media. Give an example.	02
1	3.	Explain the leveling effect of a solvent with an example.	02
1	4.	Strong oxidizing agents are unable to exist in liquid ammonia. Give reason.	02
1	5.	Why does liquid sulfur dioxide leads to formation of ion-pair and ion-triplet formation. Give examples	02
1	6.	Comment on the stability of silver oxide, on the basis of HSAB concept.	02
1	7.	What is super acid? Give examples.	02
1	8.	Predict the possible structures of BrF_3	02
1	9.	Write the auto-ionization of N_2O_4 and CH_3COOH.	02
1	10.	Taking the example of an aprotic solvent, explain self-ionization and neutralization reaction.	02
1	11.	Define superconductivity. Give an example of high temperature superconductor.	02
1	12.	Reason out, why does liquid SO_2 leads to ion-pair and ion-triplet formation.	02
1	13.	What are Lux-Flood acids and bases? Give examples.	02
1	14.	What is meant by lelling effect of solvents?	02
1	15.	What are high temperature super conductors? Give examples.	02
1	16.	What are supercritical fluids? Mention any two applications.	02
1	17.	What are Lux-Flood acids and bases? Give examples.	02
1	18.	What is meant by levelling effect of solvents?	02
1	19.	What are high temperature super conductors? Give examples.	02
1	20.	What are supercritical fluids? Mention any two applications.	02
2	21.	Write the structure of $[Fe(C_5H_5)_2(CO)_2]$ according to 18 electron rule.	02
2	22.	Explain the structure of zeise's salt? What are the parameters to classify the alkene complex as Dewar-chatt- Duncanson type or metllocyclopropanetype.	02
2	23.	Write a note on synthesis of metal alkene complexes and	02
2	24.	As a consequence of back bonding what is the impact on ethylene c-c bond length and bond order?	02
2	25.	What are pi acid ligands? Give examples.	02
2	26.	What are metallocenes? Give four examples.	02

2	27.	Write the molecular formula of ferrocene and comment on the possible haptacities with the structure.	02
2	28.	Comment on the hybridization of metallocyclopropane.	02
2	29.	Comment on the hybridation and bondangle of metallocyclopropane.	02
2	30.	What is 18 electron rule mention its significance?	02
2	31.	Calculate the electron count of $[PtCl]_4^{2-}$ by both neutral atom and oxidation state methods.	02
2	32.	Calculate the electron count of $HMn(CO)_5$ by both neutral atom and oxidation state methods.	02
2	33.	Comment on the stability of square planar complexes? Give example.	02
2	34.	Comment on the exceptional stability of 16 electron species and given an example.	02
2	35.	Calculate the electron count of Vaska's complex with its structure?	02
2	36.	Draw the structure of $Fe_2(CO)_9$ and assign IR stretching frequency of terminal and bridged carbonyl ligand.	02
2	37.	Identify the first row transition metal for $[M(CO)_3(PPh_3)]^-$	02
2	38.	Identify the first row transition metal for $HMn(CO)_5$	02
2	39.	Identify the first row transition metal for $(\eta\text{-}C_8H_8)M(CO)_3$	02
2	40.	Which is the stable metallocene among ferrocene, nicklocene and cobaltocen? Comment.	02
2	41.	What are metallocenes? Give examples.	02
2	42.	Predict which of the complexes [V(CO)6]-, Cr(CO)6 and [Mn(CO)6]+ has the shortest C-O bond.	02
2	43.	What are metal carbonyl compounds?	02
2	44.	Why are cobaltocene and nickelocene less stable than ferrocene?	02
2	45.	Comment on the aromaticity of ferrocene complex.	02
2	46.	Show the electron count for ferrocene molecule according to 18 electron and write its electronic configuration.	02
2	47.	Mention two synthesis of ferrocene.	02
2	48.	Give two examples for synthesis of metal nitrosyl complexes.	02
2	49.	Give two examples for synthesis of metal carbonyl complexes.	02
2	50.	Mention two examples for synthesis of metal carbonyl complexes.	02
2	51.	Give one example with structure of metal nitrosyl complex containing both linear and bent nitrosyl ligand.	02
2	52.	Give the IUPAC name of ferrocene and also mention the electron count?	02
2	53.	What is the important consequence of π backbonding?	02
2	54.	Which of the two isoelectronic compounds [Cr(CO)6] and [V(CO)6]- will have higher CO stretching frequency? Why?	02
2	55.	Which of the two chromium compounds $[Cr(CO)_5(PEt_3)]$ and $[Cr(CO)_5(PPh_3)]$ will have lower CO stretching frequency? Why?	02

2	56.	Which of the two chromium compounds $[Cr(CO)_5(PEt_3)]$ and $[Cr(CO)_5(PPh_3)]$ will have lower CO stretching frequency? Which will have a shorter M-C bond?	02
2	57.	The V-C bond distances in $V(CO)_6^-$ and $V(CO)_6$ are 1.93 and 2.00 Å. Justify the difference in bond distances in these compounds.	02
2	58.	How does back donation alter the C-C bond distances in metal-alkene complexes?	02
2	59.	How are linear and bent nitrosyl ligands distinguished by IR spectroscopy?	02
2	60.	All metallocenes are reactive towards electrohilic reagents than benzene. Give reason.	02
2	61.	Give two preparations for metal allyl complexes.	02
2	62.	Give two preparations for metal alkene complexes.	02
2	63.	Give two preparations for metal alkyne complexes.	02
2	64.	Give two preparations for metal arene complexes.	02
2	65.	Give reasons which illustrates the aromaticity of ferrocene.	02
2	66.	Write the structure of ferrocene and state whether it is diamagnetic or paramagnetic.	02
2	67.	What is synergic effect? How does it correlate the metal carbonyl bonding?	02
2	68.	Define 18 electron rule with example.	02
2	69.	Deduce the electron count in $H_2Fe(CO)_4$.	02
2	70.	What are pi-acceptor ligands? Discuss in detail the nature of bonding involved in metal carbonyls.	02
3	71.	Which of the following are having regular geometry? CCl_4 b. NH_3 c. PCl_5 d. CH_4	02
3	72.	Which of the following molecules are having distorted geometry? CCl_4 b. CH_3Cl c. NH_3 d. PH_3	02
3	73.	What is the geometry of ICl_2^- and TeF_5^- ?	02
3	74.	Among the following which molecule has the planar geometry. SO_2, SO_4^{2-}, CO_3^{2-}, CO_2	02
3	75.	Comment on the shape and geometry of NO_2^+, NO_2, NO_2^-.	02
3	76.	In which of the pairs the number of lone pairs are identical XeO_4, ICl_4- XeO_2F_2, ICl_4- XeO_4, ClF_3 XeF_4, ClF_3	02
3	77.	What is the shape of ClF_4^- and ClF_2^- with and without the lone pair of electrons?	02
3	78.	Draw the structure and mention the hybridization of the ClF_4^- and ClF_2^-	02
3	79.	XeF2 is isostructure with	02
3	80.	Give an example of isostructure for the XeF2 molecule?	02

3	81.	Molecular shape of CF4, SF4 and XeF4 The same with 2, 0, 1 lone pair of electrons The same with 1, 1, 1 lone pair of electrons Different with 0, 1, 2 lone pair of electrons Different with 1, 0, 2 lone pair of electrons	02
3	82.	PCl5 has the shape of trigonalbipyramidal while IF_5 has square pyramidal, Comment.	02
3	83.	Mention the bond order and magnetic property of O_2^+ and O_2.	02
3	84.	Arrange the nitrogen species in terms of increasing bond order? N_2^-, N_2, N_2^{2-}	
3	85.	Which is the most stable and why? N_2^-, N_2, N_2^{2-}	02
3	86.	Which is having the highest bond order among oxygen, superoxide and peroxide molecule.	02
3	87.	Draw the energy level diagram of HCl and calculate the bond order.	02
3	88.	Draw the energy level diagram of HF and mention the bond order.	02
3	89.	Predict the possible structures of SF4	02
3	90.	Why is diborane an electron deficient comound?	02
3	91.	The STYX number of B5H9 is 5120. How many B-H-B bonds are present in it?	02
3	92.	Classify the following boranes by structural type- B10H14 ; B2H7- ; B8H6.	02
3	93.	Why diborane is considered as electron deficient compound?	
3	94.	Using Wade's rule rationalize why $[B_6H_6]^{2-}$ adopts an octahedral Cage?	02
3	95.	Which one among the following CO_3^{2-}, SO_3, XeO_3 and NO_3^- have planar structure?	02
3	96.	Classify B10H14 as closo, nido, arachno. Using Wades rule determine the number of framework electron pairs.	02
3	97.	With an example define an electron deficient compound.	02
1	98.	Distinguish between protic and aprotic solvents. Discuss the usefulness of anhydrous sulfuric acid as a reaction medium.	05
1	99.	State the Usanovich concepts of acids and bases and explain its importance in high temperature inorganic reactions.	05
1	100.	Citing examples, discuss the acid base and complex formation in liquid SO_2. Reason out, why does liquid SO_2 leads to ion-pair and ion-triplet formation.	05
1	101.	What conspicuous properties of ammonia have been considered of significance importance for its choice as a useful solvent? Discuss the nature and applications of metal ammonia solution.	05
1	102.	Explain Lux-Flood concept of acids and bases with examples. Write a note on leveling effect of solvents.	05
1	103.	Discuss the Pearson's HSAB concept of acid and base.	05

1	104.	Outline the different types of reactions studied in anhydrous sulfuric acid. What are the limitations of this solvents?	05
1	105.	Explain the variations in conductance and magnetic properties of solutions of alkali metals in liquid ammonia.	05
	106.	How is LiAlH4 synthesized? Discuss its importance in synthetic chemistry.	05
	107.	Discuss the Pearsons's concept of hard and soft acids and bases. Explain the application of the concept with examples.	05
1	108.	Discuss the reactions carried out using N2O4 and HF as solvents.	05
1	109.	Explain the leveling effect of a solvent with an example.	05
1	110.	Write self ionization reaction of N2O4 and substantiate the ionization with two reactions.	05
1	111.	Will Cu^{2+} react more strongly with OH^- or NH_3? With O^{2-} or S^{2-} ? Explain this on the basis of HSAB concept.	05
1	112.	Explain the solvent system definition of acids and bases using an acid base neutralization reaction in HF.	05
1	113.	Explain Lux-Flood and Usanovich concept.	05
1	114.	How does liquid ammonia reacts with metal explain with suitable example.	05
1	115.	With suitable example explain acid-base reactions and solvolysis reactions take place in presence of liquid N2O4.	05
1	116.	What is hard acid? Explain the application of HSAB concept.	05
1	117.	Explain the complex formation reactions and reduction-oxidation reactions takes place in the presence of liquid sulfur dioxide.	05
1	118.	With suitable examples explain the classification of organometallic compounds on the basis of their bond type.	05
1	119.	Explain the preparation and bonding of manganese carbonyls.	05
1	120.	Describe the bonding and structures of iron and nickel organometallic complexes.	05
1	121.	Describe the structure and bonding in metal complexes containing arene and allyl ligands.	05
1	122.	Discuss the significance of HF and BrF3 as non-aqueous solvents.	05
1	123.	Outline the different types of reactions studied in anhydrous sulfuric acid. What are the limitations of this solvents?	05
2	124.	Write the Classification of Organometallic compounds?	05
2	125.	Briefly explain the classification of organometallic compounds?	05
2	126.	Discuss on bonding of metal carbonyl complexes?	05
2	127.	Explain the bonding involved in metal carbonyl complexes?	05
2	128.	Discuss on bonding of metal carbonyl complexes? How vibrational spectroscopy helps in structural elucidation of metal carbonyl?	05

2	129.	How infrared spectroscopy helps in structural elucidation of metal carbonyl? Comment on the variation of C-O stretching frequency of the following complexes Ni(CO)4, 2060 cm-1 ; $[Co(CO)_4]^-$,1890cm^{-1}; $[Fe(CO)_4]^{2-}$, 1790 cm^{-1}	05
2	130.	Explain the variation of carbon-oxygen stretching frequency in the following carbonyl compounds$[Mn(CO)_6]^+$, 2090 cm^{-1}; $[Cr(CO)_6]$, 2000 cm^{-1}; and $[V(CO)6]^-$, 1860cm^{-1}	05
2	131.	Write on the various methods of synthesis of metal carbonyls compounds.	05
2	132.	Explain the structure of zeise's salt? What are the parameters to classify the alken complex as Dewar-chatt- Duncanson type or metllocyclopropanetype?	05
2	133.	Write a note on synthesis and the structure of zeise's salt.	05
2	134.	Explain the bonding of metal-alkene complex based on as Dewar-chatt-Duncanson model.	05
2	135.	How are metal nitrosyl prepared? Discuss with examples the structure of linear and bent nitrosyls.	05
2	136.	Discuss the methods of preparation and structures of metal nitrosyls.	05
2	137.	Outline the synthesis and bonding in metal-allyl complexes.	05
2	138.	State 18 electron rule in organometllics. Mention its limitations.	05
2	139.	Outline the synthesis and bonding in metal-alkene complexes.	05
2	140.	Outline the synthesis and bonding in metal-arene complexes.	05
2	141.	Discuss the different methods of preparations of metal nitrosyls. Give an example for linear and bent nitrosyl.	
2	142.	Explain the bonding involved in ferrocene using MO concept.	05
2	143.	Discuss the classification of organometallic compound on the basis of the bond type.	
2	144.	Explain the structure of the zeise's salt and bonding involved in it.	
2	145.	Explain the structure and bonding in metal alkyne complexes.	05
2	146.	Give an account of 18 electron rule in organometallics. What are its limitations?	05
2	147.	Briefly explain on M-CO bonding and explain the usefulness of infrared strudies in it.	05
2	148.	Outline the preparation of $Mn_2(CO)_{10}$ and $Co(CO)_8$. Explain the structure and bonding in metal carbonyls.	05
2	149.	Illustrate the preparation and mode of bonding in metal allyl complexes.	05
2	150.	Discuss any two methods for the preparation of ferrocene. Explain its structure.	05
2	151.	Discuss the preparation, structure and bonding in metal arene complexes.	05
2	152.	Illustrate the preparation and mode of bonding in metal nitrosyls.	05
2	153.	Explain the preparation, structure and bonding in metal containing alkyne ligands.	05

3	154.	Draw the Lewis structure for the following molecules and predict the molecular geometry with and without non bonded electrons. XeF_2, XeF_4, XeO_3, $XeOF_4$	05
3	155.	Draw the Lewis structure for the following molecules and predict the molecular geometry. $SnBr_4$, TeF_6, XeO_4, ClF_3	05
3	156.	Draw the walsh diagram of bent and linear triatomic molecule.	05
3	157.	Draw the walsh and MO diagram of bent and linear BH_2 molecule.	05
3	158.	Draw the molecular orbital diagram of carbonate ion and comment on the magnetic property.	05
3	159.	Draw the enegy level diagram of carbonate ion involving delocalized pi bonding.	05
3	160.	Sketh the energy level diagram of planar trigonal molecule involving delocalized pi bonding and comment on the magnetic property	05
3	161.	Sketch the energy level diagram of linear triatomic molecule involving delocalized pi bonding and comment on the magnetic property	05
3	162.	Sketch the energy level diagram of carbondioxide molecule involving delocalized pi bonding and comment on the magnetic property	05
3	163.	Draw the molecular orbital diagram of nitrate ion and comment on the magnetic property.	05
3	164.	Draw the molecular orbital diagram of oxygen molecule and comment on the bond order, stability and magnetic behavior of O_2, O_2^- O_2^{2-} and O_2^+.	05
3	165.	Draw the molecular orbital diagram of nitrogen molecule and comment on the bond order, stability and magnetic behavior of dinitrogencation, dinitrogen anion and nitrogen molecule.	05
3	166.	Sketch the energy level diagram for CO and NO molecule.	05
3	167.	N_2 and O_2 have different bonding and magnetic properties. Explain with the help of M.O diagrams.	05
3	168.	Discuss the structure and bonding involved in diborane.	05
3	169.	Draw the MO energy level diagram of N_2 and CO and calculate the bond order and predict the magnetic behavior.	05
3	170.	How do bond lengths vary in the dioxygen species of O_2, O_2^+, O_2^-, O_2^{2-} ? Which of these would be paramagnetic? Explain.	05
3	171.	Verify whether $(B_{12}H_{12})^{2-}$ and $C_2B_{10}H_{12}$ are isoelectronic. Using Wade's rules show that these molecules belong to closo type of boranes.	05
3	172.	What is Walsh diagram? Discuss the MO correlation diagram for triatomic molecules.	05
3	173.	Construct the MO diagram for carbonate ion and discuss its structure.	05
3	174.	What is STYX number? Calculate Styx number and draw valence structure for $(B5H5)^{2-}$	05

3	175.	Based on the VSEPR model account for the structures of AB2E , AB2E2, AB2E3 type of molecules? Mention the drawbacks of VSEPR model.	05
3	176.	List out Wade's rules and explain two methods of preparation of polyhedral boranes.	05
3	177.	With near sketch of molecular orbital diagram, explain how and why the molecular orbitals of N_3^- differ from those of CO_2	05
3	178.	Based on VSEPR theory, explain the structure of SF_4 and ClF_3	05
3	179.	Depict the Walsh diagram for AH2 molecule and based on it explain the shapes of linear and bent triatomic molecules.	05
3	180.	Set up MO diagram for nitrogen molecule. Using the diagram explain the relative stabilities of N_2^+, N_2^-, N_2	05
3	181.	How diborane is prepared? Explain the structure and bonding in diborane.	05
3	182.	Construct the MO energy level diagram for NO^{3-} involving delocalized pi bonding? Point out the salient feature of the diagram	05
3	183.	Verify whether $(B_{12}H_{12})^{2-}$ and $C_2B_{10}H_{12}$ are isoelectronic. Using Wade's rules show that these molecules belong to closo type of boranes.	05
3	184.	With near sketch of molecular orbital diagram, explain how and why the molecular orbitals of N_3^- differ from those of CO_2	05

Question Paper Pattern- Model Question Paper

Q.P Code: MCH20A102

ST. PHILOMENA'S COLLEGE (AUTONOMOUS), MYSORE			
I SEMESTER –COURSE M.SC FINAL EXAMINATION			
SUBJECT : CHEMISTRY			
TITLE: INORGANIC CHEMISTRY -I (HC)			
TIME: 3 HRS			**MAX MARKS: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	What is Walsh diagram? Give examples.	2
	b	Arrange the nitrogen species in terms of increasing bond order? N_2^-, N_2, N_2^{2-}	2
	c	Predict the possible structures of SF_4	2
	d	Which of the following molecules are having distorted geometry? CCl_4 b. CH_3Cl c. NH_3 d. PH_3	2
	e	Define metalloboranes. To which structure does $B_4H_6(CoCp)_2$ belongs.	2
	f	Draw the energy level diagram of HCl and calculate the bond order.	2
	g	Identify the first row transition metal for $[M(CO)_3(PPh_3)]^-$	2
	h	Comment on the exceptional stability of 16 electron species and given an example.	2
	i	What are metal carbonyl compounds? How are they classified?	2
	j	What are acids bases according to Lux-Flood concept.	2
	k	Write the auto-ionization of N_2O_4 and CH_3COOH.	2
	l	What is super acid? Give examples.	2
PART-B			
		Answer any Five of the following:	5 x 10= 50
2	a	Discuss on bonding of metal carbonyl complexes?	5
	b	How does infrared spectroscopy helps in structural elucidation of metal carbonyl? Comment on the variation of C-O stretching frequency of the following complexes Ni(CO)4, 2060 cm-1 ; $[Co(CO)_4]^-$,1890cm^{-1}; $[Fe(CO)_4]^{2-}$, 1790 cm^{-1}	5
3	a	With suitable example explain acid-base reactions and solvolysis reactions take place in presence of liquid N2O4.	5
	b	Discuss the reactions carried out using N2O4 and HF as solvents.	5
4	a	Explain the bonding involved in ferrocene using MO concept.	5

	b	Explain Lux-Flood concept of acids and bases with examples. Write a note on leveling effect of solvents.	5
5	a	Discuss the Pearson's HSAB concept of acid and base.	5
	b	How are metal nitrosyl prepared? Discuss with examples the structure of linear and bent nitrosyls.	5
6	a	Draw the Lewis structure for the following molecules and predict the molecular geometry. $SnBr_4$, TeF_6, XeO_4, ClF_3	5
	b	Verify whether $(B_{12}H_{12})^{2-}$ and $C_2B_{10}H_{12}$ are isoelectronic. Using Wade's rules show that these molecules belong to closo type of boranes.	5
7	a	State 18 electron rule in organometllics. Mention its limitations.	5
	b	Sketch the energy level diagram of linear triatomic molecule involving delocalized pi bonding and comment on the magnetic property	5
8	a	Set up MO diagram for nitrogen molecule. Using the diagram explain the relative stabilities of N_2^+, N_2^-, N_2	5
	b	List out Wade's rules and explain two methods of preparation of boranes and polyhedral boranes.	5